

Exercise 3: Stored Procedures

Scenario

This exercise demonstrates the creation of PL/SQL stored procedures to automate common banking operations such as monthly interest calculation, employee bonus updates, and fund transfers.

Objective

The objective of this exercise is to create reusable PL/SQL stored procedures that simplify banking operations, improve code reusability, reduce redundancy, and enhance database management.

1. Understanding Stored Procedures

A stored procedure is a named PL/SQL block that is stored in the database and can be executed whenever required. It helps reduce code duplication, improves performance, and centralizes business logic.

Benefits of Stored Procedures:

- Improves code reusability
- Reduces redundancy
- Enhances database security
- Improves performance
- Simplifies maintenance

2. Implementation

Scenario 1: Process Monthly Interest

This procedure calculates and updates monthly interest for all savings accounts by increasing the account balance by 1%.

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
```

```
IS
```

```
BEGIN
```

```
    UPDATE Accounts
```

```
    SET Balance = Balance + (Balance * 0.01)
```

```
    WHERE AccountType = 'Savings';
```

```
    COMMIT;
```

```
END;
```

```
/
```

Scenario 2: Update Employee Bonus

This procedure updates the salary of employees belonging to a specific department by applying a bonus percentage.

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
```

```
    p_department VARCHAR2,
```

```
    p_bonus NUMBER
```

```
)
```

```
IS
```

```
BEGIN
```

```
    UPDATE Employees
```

```
    SET Salary = Salary + (Salary * p_bonus / 100)
```

```
    WHERE Department = p_department;
```

```
    COMMIT;
```

```
END;
```

```
/
```

Scenario 3: Transfer Funds

This procedure transfers money from one account to another after checking whether sufficient balance is available.

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_from NUMBER,
```

```
    p_to NUMBER,
```

```
    p_amount NUMBER
```

```
)
```

```
IS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    SELECT Balance
```

```
    INTO v_balance
```

```
    FROM Accounts
```

```
    WHERE AccountID = p_from;
```

```
IF v_balance >= p_amount THEN
```

```
    UPDATE Accounts
```

```
    SET Balance = Balance - p_amount
```

```
    WHERE AccountID = p_from;
```

```
    UPDATE Accounts
```

```
    SET Balance = Balance + p_amount
```

```
    WHERE AccountID = p_to;
```

```
COMMIT;
```

```
DBMS_OUTPUT.PUT_LINE('Transfer Successful');
```

```
ELSE
```

```
    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
```

```
END IF;
```

```
END;
```

```
/
```

3. Output

Expected outputs:

- Procedure ProcessMonthlyInterest created successfully.
- Procedure UpdateEmployeeBonus created successfully.
- Procedure TransferFunds created successfully.
- Transfer Successful (or Insufficient Balance based on the account balance).

4. Conclusion

Successfully implemented PL/SQL stored procedures for processing monthly interest, updating employee bonuses, and transferring funds between accounts. These procedures

demonstrate modular programming, improve code reusability, simplify database operations, and enhance the efficiency of banking applications.

Exercise 1: Control Structures

Scenario

This exercise demonstrates the use of PL/SQL control structures to perform common banking operations such as applying loan discounts, promoting customers to VIP status, and sending loan repayment reminders.

Objective

The objective of this exercise is to understand and implement PL/SQL control structures such as loops, conditional statements, and cursors to automate banking operations efficiently.

1. Understanding Control Structures

Control structures in PL/SQL are used to control the flow of program execution. They help perform repetitive tasks, make decisions, and process multiple records efficiently.

Common control structures include:

- IF...THEN...ELSE statements
- LOOP, FOR LOOP, and WHILE LOOP
- CURSORS for processing multiple rows

2. Implementation

Scenario 1: Apply Discount to Loan Interest Rates

This PL/SQL block loops through all customers and applies a 1% discount to loan interest rates for customers who are above 60 years of age.

```
DECLARE
```

```
CURSOR c_customers IS
```

```
SELECT CustomerID, Age FROM Customers;
```

```
BEGIN
```

```
FOR customer IN c_customers LOOP
```

```
IF customer.Age > 60 THEN
```

```
UPDATE Loans
```

```
SET InterestRate = InterestRate - 1
```

```
WHERE CustomerID = customer.CustomerID;
```

```
END IF;
```

```
END LOOP;  
COMMIT;  
END;  
/
```

Scenario 2: Promote Customers to VIP Status

This PL/SQL block checks customer balances and marks customers as VIP if their balance is greater than 10,000.

```
DECLARE  
CURSOR c_customers IS  
SELECT CustomerID, Balance FROM Customers;  
BEGIN  
FOR customer IN c_customers LOOP  
IF customer.Balance > 10000 THEN  
UPDATE Customers  
SET IsVIP = 'TRUE'  
WHERE CustomerID = customer.CustomerID;  
END IF;  
END LOOP;  
COMMIT;  
END;  
/
```

Scenario 3: Send Loan Due Reminders

This PL/SQL block retrieves all loans due within the next 30 days and displays a reminder message for each customer.

```
DECLARE  
CURSOR c_loans IS  
SELECT CustomerID, DueDate  
FROM Loans  
WHERE DueDate BETWEEN SYSDATE AND SYSDATE + 30;  
BEGIN  
FOR loan IN c_loans LOOP
```

```
DBMS_OUTPUT.PUT_LINE('Reminder: Loan due for Customer ID ' || loan.CustomerID || '
on ' || loan.DueDate);
```

```
END LOOP;
```

```
END;
```

```
/
```

3. Output

Expected outputs:

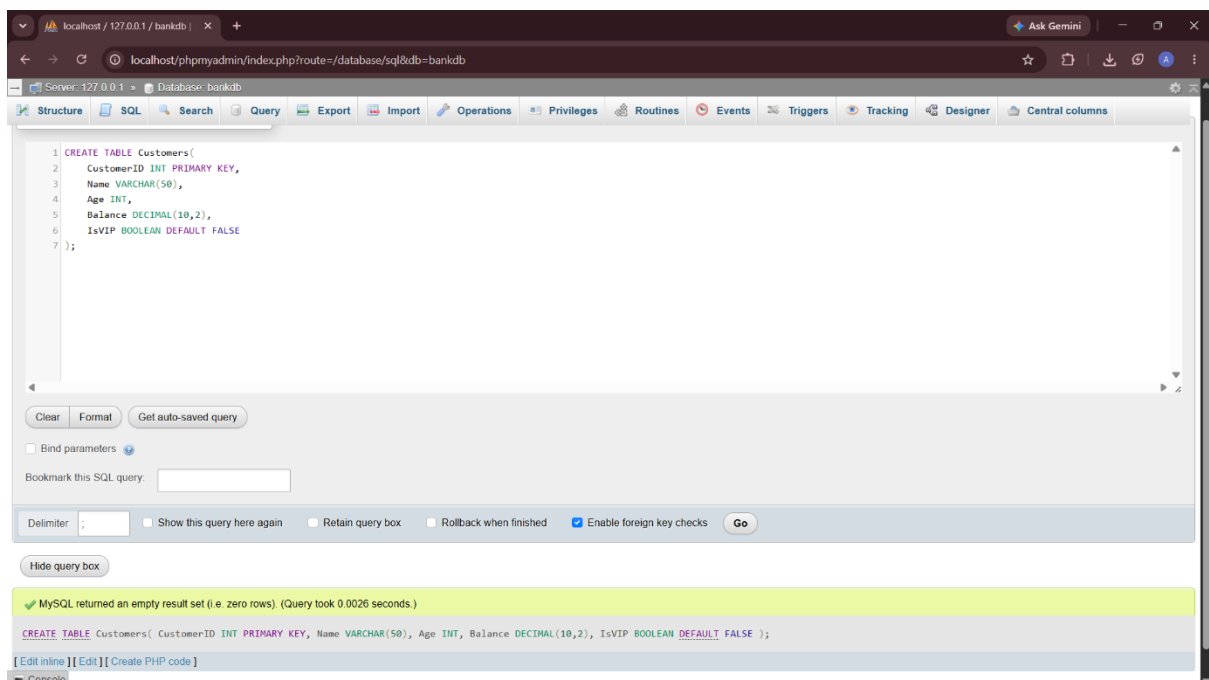
- Interest rates updated successfully for eligible customers.
- Customers with balance above 10,000 promoted to VIP status.
- Reminder messages displayed for customers whose loan due dates are within the next 30 days.

4. Conclusion

Successfully implemented PL/SQL control structures to automate banking operations. Using loops, conditional statements, and cursors improved code readability, reduced manual effort, and demonstrated efficient database programming techniques.

Output screenshots

Exercise 1



localhost / 127.0.0.1 / bankdb | x + Ask Gemini

localhost/phpmyadmin/index.php?route=/database/sql&db=bankdb

Server: 127.0.0.1 | Database: bankdb | Table: Customers

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Clear Format Get auto-saved query

☐ Bind parameters

Bookmark this SQL query:

Delimiter: ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks

Hide query box.

Showing rows 0 - 2 (3 total, Query took 0.0008 seconds.)

SELECT * FROM Customers;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CustomerID	Name	Age	Balance	IsVIP
☐ Edit Copy Delete	1	Anusha	85	15000.00	0
☐ Edit Copy Delete	2	Rahul	45	9000.00	0
☐ Edit Copy Delete	3	Priya	78	20000.00	0

☐ Check all With selected:

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Console

localhost / 127.0.0.1 / bankdb | x + Ask Gemini

localhost/phpmyadmin/index.php?route=/table/s...

Server: 127.0.0.1 | Database: bankdb | Table: Loans

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Show query box.

Showing rows 0 - 2 (3 total, Query took 0.0011 seconds.)

SELECT * FROM Loans;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	LoanID	CustomerID	InterestRate	DueDate
☐ Edit Copy Delete	101	1	10.50	2026-07-10
☐ Edit Copy Delete	102	2	11.00	2026-08-15
☐ Edit Copy Delete	103	3	9.50	2026-07-20

☐ Check all With selected:

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/s...

Server: 127.0.0.1 » Database: bankdb » Table: Loans

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

Mo

Run SQL query/queries on table bankdb.Loans:

1 UPDATE Loans l

2 JOIN Customers c

3 ON l.CustomerID=c.CustomerID

4 SET l.InterestRate=l.InterestRate-1

5 WHERE c.Age>60;

LoanID

CustomerID

InterestRate

DueDate

SELECT *

SELECT

INSERT

UPDATE

DELETE

Clear

Format

Get auto-saved query

<<

☐ Bind parameters

Bookmark this SQL query:

Delimiter ;

☐ Show this query here again

☐ Retain query box

☐ Rollback when finished

☒ Enable foreign key checks

Go

Hide query box

✓ 2 rows affected. (Query took 0.0008 seconds.)

UPDATE Loans l JOIN Customers c ON l.CustomerID=c.CustomerID SET l.InterestRate=l.InterestRate-1 WHERE c.Age>60;

[Edit inline] [Edit] [Create PHP code]

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/s...

Server: 127.0.0.1 > Database: bankdb > Table: Customers

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsMore

Run SQL query/queries on table bankdb.Loans:

1 UPDATE Customers

2 SET IsVIP=TRUE

3 WHERE Balance>10000;

4

5

6 SELECT * FROM Customers;

LoanID

CustomerID

InterestRate

DueDate

SELECT *SELECTINSERTUPDATEDELETEClearFormatGet auto-saved query<<

☐ Bind parameters

Bookmark this SQL query:

Delimiter ;

☐ Show this query here again

☐ Retain query box

☐ Rollback when finished

☒ Enable foreign key checks

Simulate queryGo

Hide query box

✓ 2 rows affected. (Query took 0.0014 seconds.)

UPDATE Customers SET IsVIP=TRUE WHERE Balance>10000;

[Edit inline][Edit][Create PHP code]

✓ Showing rows 0 - 2 (3 total, Query took 0.0013 seconds.)

SELECT * FROM Customers;

☐ Profiling [Edit inline][Edit][Explain SQL][Create PHP code][Refresh]

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/s...

Server: 127.0.0.1 » Database: bankdb » Table: Customers

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsMore

☐ Bind parameters

Bookmark this SQL query:

Delimiter: ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks

Simulate queryGo

Hide query box

✓ 2 rows affected. (Query took 0.0014 seconds.)

UPDATE Customers SET IsVIP=TRUE WHERE Balance>10000;

[Edit inline][Edit][Create PHP code]

✓ Showing rows 0 - 2 (3 total, Query took 0.0013 seconds.)

SELECT * FROM Customers;

☐ Profiling [Edit inline][Edit][Explain SQL][Create PHP code][Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

CustomerIDNameAgeBalanceIsVIP

☐ Edit Copy Delete

1Anusha6515000.001

☐ Edit Copy Delete

2Rahul459000.000

☐ Edit Copy Delete

3Priya7020000.001

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Bookmark this SQL query

☐ Let every user access this bookmark

Console

localhost / 127.0.0.1 / bankdb / x + Ask Gemini

localhost/phpmyadmin/index.php?route=/table/s...

Server: 127.0.0.1 » Database: bankdb » Table: Loans

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking

```

4 1.DueDate
5 FROM Customers c
6 JOIN Loans 1
7 ON c.CustomerID=1.CustomerID
8 WHERE 1.DueDate
9 BETWEEN CURDATE()
10 AND DATE_ADD(CURDATE(),INTERVAL 30 DAY);

```

DueDate

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters ⓘ

Bookmark this SQL query:

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✓ Showing rows 0 - 1 (2 total, Query took 0.0045 seconds.)

```

SELECT c.Name, 1.LoanID, 1.DueDate FROM Customers c JOIN Loans 1 ON c.CustomerID=1.CustomerID WHERE 1.DueDate BETWEEN CURDATE() AND
DATE_ADD(CURDATE(),INTERVAL 30 DAY);

```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

Name	LoanID	DueDate
Anusha	101	2026-07-10
Priya	103	2026-07-20

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Console

Exercise 3

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Accounts

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingMore

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

SELECT * FROM Accounts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

AccountIDCustomerNameAccountTypeBalance

☐

Edit

Copy

Delete

101

Anusha

Savings

10000.00

☐

Edit

Copy

Delete

102

Rahul

Savings

15000.00

☐

Edit

Copy

Delete

103

Priya

Current

20000.00

☐

Check all

With selected:

Edit

Copy

Delete

Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Accounts

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingMo

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

SELECT * FROM Accounts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

AccountIDCustomerNameAccountTypeBalance

☐

EditCopyDelete

101AnushaSavings10201.00

☐

EditCopyDelete

102RahulSavings15301.50

☐

EditCopyDelete

103PriyaCurrent20000.00

Check all

With selected:

EditCopyDeleteExport

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 > Database: bankdb > Table: Employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

More

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

SELECT * FROM Employees;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

EmployeeID

EmployeeName

Department

Salary

☐

Edit

Copy

Delete

1

John

IT

55000.00

☐

Edit

Copy

Delete

2

Alice

HR

45000.00

☐

Edit

Copy

Delete

3

David

IT

66000.00

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Accounts

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

More

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0252 seconds.)

CALL UpdateEmployeeBonus('IT',10);

[Edit inline]

[Edit]

[Create PHP code]

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Employees

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingMore

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)

SELECT * FROM Employees;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmployeeIDEmployeeNameDepartmentSalary

☐	Edit Copy Delete	1 John	IT	60500.00
☐	Edit Copy Delete	2 Alice	HR	45000.00
☐	Edit Copy Delete	3 David	IT	72600.00

☐ Check all With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Accounts

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingMore

Show query box

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

SELECT * FROM Accounts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

AccountIDCustomerNameBalance

☐	Edit	Copy	Delete	101	Anusha	8000.00
☐	Edit	Copy	Delete	102	Rahul	10000.00

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Bookmark this SQL query

Label:

☐ Let every user access this bookmark

Bookmark this SQL query

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 » Database: bankdb » Table: Employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

Mor

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0088 seconds.)

CALL

TransferFunds(101,102,2000);

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Console

localhost / 127.0.0.1 / bankdb /

Ask Gemini

localhost/phpmyadmin/index.php?route=/table/sql&db...

Server: 127.0.0.1 > Database: bankdb > Table: Accounts

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

More

Show query box

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

SELECT * FROM Accounts;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

AccountID

CustomerName

Balance

Edit

Copy

Delete

101

Anusha

6000.00

Edit

Copy

Delete

102

Rahul

12000.00

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Console